Understanding Marketplace Performance Across Geographies: Socioeconomic Trends and Future Considerations

Capstone Project Proposal:

Methodology

San Francisco State University

MS Business Analytics · BUS 895

Gregory Joshua

Research Questions and Approach

To explore and answer the questions:

1. What socioeconomic and market characteristics across geographies are associated with Marketplace performance?
2. Which areas currently exhibit these characteristics and outcomes?
3. How are these socioeconomic and market characteristics expected to change over the next five to ten years?
4. What considerations for future market analysis emerge from these findings?

This project will begin by exploring publicly available data from Covered California, the Centers for Medicare & Medicaid Services (CMS), the U.S. Census Bureau, and the Bureau of Labor Statistics (BLS).

Relevant datasets will then be extracted, cleaned, standardized, and integrated into a unified analytical dataset, allowing socioeconomic characteristics, market conditions, and Marketplace performance to be compared across geographic areas and time periods.

Once the data is prepared, exploratory analysis and statistical modeling will be used to identify relationships between socioeconomic and market characteristics and Marketplace performance, as well as geographic areas that currently exhibit these characteristics and outcomes. Historical trends and available projections will then be examined to estimate how relevant characteristics may change over the next five to ten years.

Finally, the findings will be synthesized to identify considerations for future Marketplace analysis. The following sections describe the data sources, data engineering processes, and planned analytical methods in greater detail.

Data Sources and Available Datasets

This project utilizes publicly available datasets from four primary sources: the Centers for Medicare & Medicaid Services (CMS), Covered California, the U.S. Census Bureau, and the Bureau of Labor Statistics (BLS). Together, these sources provide information on Marketplace enrollment and competition, insurance premiums, and socioeconomic characteristics across California's 58 counties. The analysis covers the years 2016 through 2026, where data is available.

Centers for Medicare & Medicaid Services (CMS)

CMS provides Qualified Health Plan (QHP) Public Use Files containing information on health insurance plans offered through the Marketplace. Three datasets are utilized: Service Area, Plan Attributes, and Rates. These provide information on geographic plan availability, participating insurance carriers, plan characteristics, and premiums. This data supports the analysis of market competition, Kaiser Permanente's geographic participation, and premium differences across counties and years.

Covered California

Covered California's Active Member Profile datasets provide information on health insurance enrollment across California's individual Marketplace. These datasets are used to measure Kaiser Permanente enrollment, total Marketplace enrollment, and Kaiser Permanente's share of enrollment within geographic markets. Enrollment and market share serve as key measures of Marketplace performance throughout the analysis.

U.S. Census Bureau

The Census Bureau's American Community Survey (ACS) provides demographic and socioeconomic estimates at the county level. The datasets utilized include total population (B01003), median household income (B19013), poverty status (S1701), and health insurance coverage (S2701). These variables provide a basis for evaluating relationships between the socioeconomic characteristics of geographic areas and Marketplace performance.

Bureau of Labor Statistics (BLS)

Two BLS datasets are utilized: Local Area Unemployment Statistics (LAUS) and Quarterly Census of Employment and Wages (QCEW). LAUS provides county-level measures of labor force participation, employment, and unemployment, while QCEW provides information on establishments, employment, and wages. Together, these datasets provide additional context on local economic conditions and labor market characteristics that may be associated with Marketplace outcomes.

Collectively, these datasets support the development of a county-year analytical dataset, allowing Marketplace performance, market characteristics, and socioeconomic indicators to be examined across geographic areas and over time. Differences in source coverage, geographic definitions, and reporting periods are addressed through the data engineering process described in the following section.

Data Engineering

To prepare the datasets for analysis, this project utilizes an Extract, Load, Transform (ELT) pipeline built with DuckDB and dbt (data build tool). DuckDB serves as the analytical database, while dbt manages SQL-based transformations, data modeling, testing, and documentation. The

pipeline integrates data from the four previously described sources into a standardized analytical structure, enabling comparisons across California's 58 counties and the years 2016–2026, where data is available.

Data Extraction and Loading

Data is obtained from publicly available files published by CMS, Covered California, the Census Bureau, and BLS. These sources provide data in varying formats, structures, and reporting periods. Source files are collected and made available for processing in DuckDB, where they can be queried and transformed using SQL.

The ELT approach allows data to be loaded before the majority of transformations are performed. This preserves source-level information while enabling cleaning, standardization, and integration to be managed within dbt.

Data Transformation and Modeling

Transformations are organized using a layered dbt architecture consisting of staging, intermediate, and mart models. Each layer serves a distinct purpose in preparing the data for analysis.

Staging models standardize the structure and content of individual source datasets. These transformations include renaming columns, assigning consistent data types, handling missing values, and standardizing geographic and temporal identifiers. Because some source schemas change between reporting years, conditional SQL logic is also used to accommodate differences in column availability and structure.

Intermediate models integrate related datasets and perform transformations necessary to establish consistent analytical relationships. For example, CMS Plan Attributes, Service Area, and Rate data are combined to associate insurance plans, carriers, geographic availability, and premiums. Reference tables maintained as dbt seeds support geographic mappings, including the relationship between counties and insurance rating areas.

Finally, mart models consolidate the transformed data into three primary analytical datasets:

1. Market: mart_cty_yr_market — insurance carrier participation, plan availability, competition, and premium characteristics.
2. Performance: mart_cty_yr_performance — Marketplace enrollment and Kaiser Permanente performance measures.
3. Socioeconomic: mart_cty_yr_socioeconomic — demographic, income, poverty, insurance coverage, and labor market characteristics.

Each mart is structured at the county-year level, using county FIPS codes and reporting years as common identifiers. This structure allows the datasets to be joined for statistical analysis while maintaining consistent geographic and temporal granularity. The overall transformation process and dependencies between dbt models are illustrated in Figure 1.

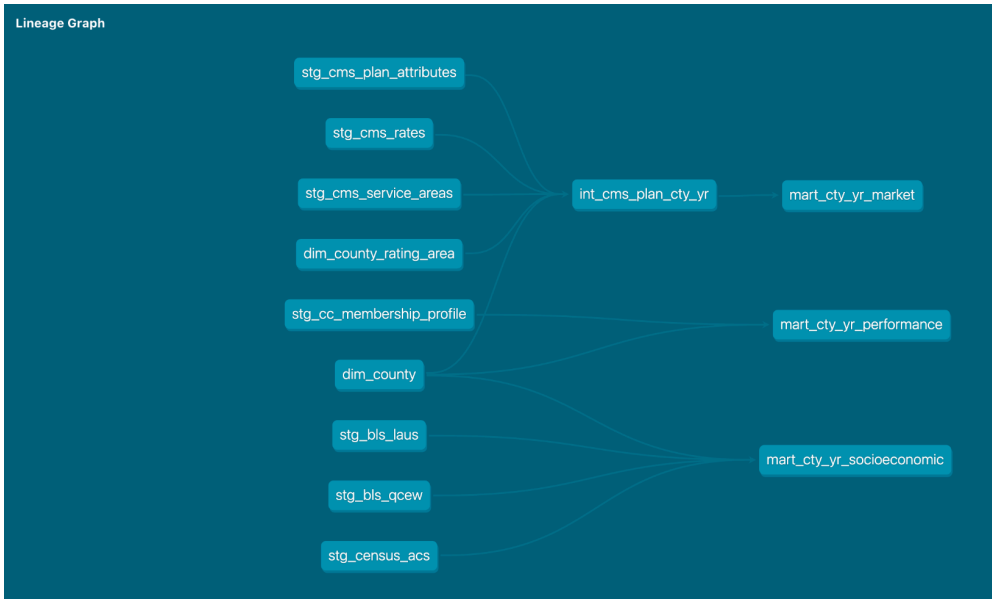


Figure 1. dbt lineage graph illustrating the transformation of source data through staging, intermediate, and mart models.

Data Validation and Quality Assurance

Data quality is evaluated using automated tests within dbt. More than 100 tests have been implemented to assess null values, uniqueness, referential integrity, required identifiers, and the expected granularity of analytical tables. These checks help identify issues such as duplicate county-year records, missing geographic identifiers, and inconsistent relationships between datasets.

Additional attention is given to limitations in the underlying public data. Differences in reporting periods, geographic coverage, and source availability may prevent certain measures from being consistently calculated across all counties and years. Such limitations are documented, and unavailable or unreliable values are retained as missing rather than replaced with unsupported estimates.

Rationale for the Data Engineering Approach

This approach is necessary because the original datasets were not designed to be analyzed together. Differences in geographic granularity, reporting periods, naming conventions, and data structures make direct comparisons difficult without additional processing.

Using DuckDB and dbt provides a reproducible workflow in which transformations are defined in SQL, dependencies are explicitly documented, and data quality is systematically tested. The layered architecture also separates source-specific cleaning from more complex integration and analytical transformations, allowing individual components to be modified without rebuilding the entire pipeline.

Ultimately, the resulting county-year analytical datasets provide a consistent foundation for exploring relationships between Marketplace performance and socioeconomic or market characteristics, as well as evaluating how these relationships and characteristics may change over time.

Planned Analysis and Predictive Modeling

The analytical phase of this project will utilize exploratory data analysis, statistical modeling, and predictive forecasting to investigate relationships between socioeconomic conditions, market characteristics, and Kaiser Permanente's Marketplace performance. The analysis will primarily examine California counties where Kaiser Permanente offers Marketplace plans, using the three county-year analytical datasets described in the previous section.

Exploratory Data Analysis

Exploratory data analysis (EDA) has begun using Python, pandas, and Matplotlib. Initial analyses examine historical Marketplace enrollment, Kaiser Permanente market share, geographic differences in performance, and relationships between performance and socioeconomic or market characteristics.

Pearson and Spearman correlation coefficients are used to evaluate relationships between variables. Preliminary analyses focus on the years 2016–2024, aggregating observations into county-level averages for periods when Kaiser Permanente offered Marketplace plans. This approach helps distinguish differences between geographic markets while reducing the influence of county size on comparisons.

Preliminary findings suggest positive associations between Kaiser Permanente market share and median household income, as well as negative associations with poverty, uninsured rates, and unemployment. Positive correlations are also observed with Silver-plan premiums, while the number of competing insurance carriers exhibits a comparatively weak relationship with market share.

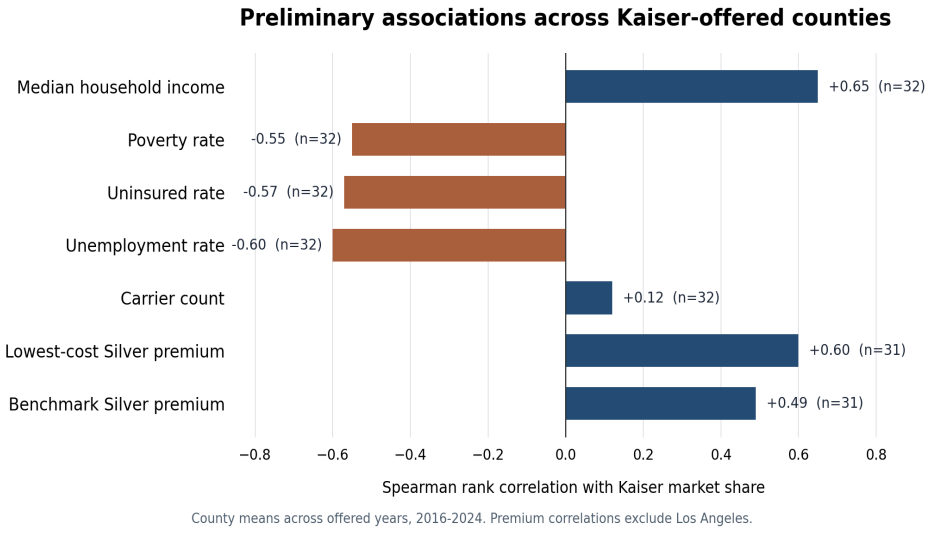


Figure 2. Spearman correlations between Kaiser Permanente market share and selected socioeconomic and market characteristics, calculated using county-level averages from 2016–2024.

Additional descriptive analysis will examine enrollment and market share trends over time, compare counties with varying performance levels, and identify geographic patterns through visualizations and maps. These findings will inform variable selection and provide context for subsequent statistical modeling.

Multiple Regression Modeling

Multiple regression will be used to estimate the relationships between socioeconomic and market characteristics and Kaiser Permanente's Marketplace performance while accounting for several factors simultaneously.

The primary dependent variable will be Kaiser Permanente's county-level Marketplace market share. A secondary analysis will examine year-over-year changes in market share, measured in percentage points, to investigate factors associated with changes in performance over time.

Independent variables under consideration include median household income, poverty rate, uninsured rate, unemployment rate, insurance carrier count, and Silver-plan premiums. These variables were selected based on their relevance to Marketplace conditions and the relationships observed during preliminary exploratory analysis.

Multiple linear regression is appropriate because it provides interpretable estimates of how each characteristic is associated with Marketplace performance while holding other included variables constant. Models incorporating county and year fixed effects will also be considered to account for persistent geographic differences and broader changes affecting all counties over time.

Model development will include examinations of multicollinearity, missing data, and potential outliers. Because certain socioeconomic measures may be closely related, predictor selection and regularization techniques may be evaluated to improve model stability. The analysis will also account for limitations in premium availability and distinguish between counties where Kaiser Permanente offers plans and those outside its service area.

Models will be evaluated using measures such as R-squared, mean absolute error (MAE), and root mean squared error (RMSE). Where feasible, historical observations will be divided chronologically into training and validation periods to assess performance on later years.

Predictive Modeling and Forecasting

To investigate how socioeconomic and market characteristics may change over the next five to ten years, historical trends will be evaluated using time-series analysis and regression-based forecasting.

Variables such as median household income, poverty rates, unemployment, and population growth will be examined individually to identify historical trends and estimate potential future values. Simple trend-based regression models will provide an initial forecasting approach, with published projections incorporated where suitable data is available.

This approach is appropriate because the project seeks to understand how the characteristics of geographic markets may evolve, rather than relying exclusively on predictions of future insurance enrollment.

Where reliable projections are unavailable, scenario analysis will be used to explore alternative future conditions. Projected socioeconomic characteristics may then be evaluated alongside the relationships identified through regression modeling to assess how the characteristics of existing markets could change.

Given the uncertainty associated with long-term economic, regulatory, and insurance market conditions, projections will be presented as estimates or scenarios rather than definitive predictions.

Geographic Analysis and Interpretation

Finally, the findings from exploratory analysis, regression modeling, and forecasting will be integrated to identify geographic areas exhibiting characteristics associated with stronger or weaker Marketplace performance.

Counties will be compared based on their current characteristics, observed Marketplace outcomes, and projected socioeconomic changes. Particular attention will be given to areas where expected changes may warrant further market investigation.

These findings will be synthesized into considerations for Kaiser Permanente's future Marketplace analysis, including geographic markets that may merit additional research and characteristics that may be useful to monitor over time.

The analysis is intended to identify statistical associations and inform future investigation, rather than establish causal relationships or directly recommend market entry or expansion decisions.